

Rooftop PV Information and Next Half-Hour NSW Demand Forecasting

A data science investigation into the forecasting value of rooftop photovoltaic (PV) information for next half-hour NSW operational demand forecasting.

RESEARCH QUESTION
To what extent does adding recent rooftop PV information improve next half-hour forecasts of NSW operational demand change compared with otherwise matched non-PV models, and when is any improvement greatest?

Field	Details
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1 Abstract

This study looks at whether including recent rooftop solar PV data can make half-hourly forecasts for NSW operational demand more accurate. To do this, we combined observed rooftop PV and demand data with existing demand forecasts, calendar variables, and daily weather observations. We used rooftop PV data spanning from 2017 to 2026 to account for long-term growth, while our main modeling dataset consisted of 36,526 aligned half-hourly records from August 2024 to September 2026.

We evaluated how well the models performed using a chronological holdout set. We compared models that included PV information against those that didn't, and we also used an external AEMO POE50 benchmark that was adjusted to match our demand-change target. The tree model enhanced with PV data reached an MAE of 62.13 MW, which was slightly better than the 63.07 MW seen in the model without PV data. For comparison, the AEMO benchmark had an MAE of 58.21 MW.

The biggest boost from adding PV data happened between 06:00 and 12:00, where the MAE dropped by 2.01 MW. Overall, the results suggest that PV information offers a modest improvement in forecast accuracy that varies depending on the time of day, though it is important to note that these results don't prove a direct causal relationship.

2 Introduction

Roughly one in three Australian homes now has rooftop solar (Albanese 2026). Since this power is generated behind the meter, more household production naturally means less electricity is pulled from the grid. Yan and Han (2023) showed this connection, proving that rooftop solar lowers daytime demand across Australia's National Electricity Market (NEM). This creates the well-known duck curve, where the midday dip in demand gets deeper and the afternoon ramp becomes much steeper as solar production drops off. As Australia moves toward more electrification, understanding these shifts is vital for forecasting grid demand.

Between 2020 and 2025, the share of Australian electricity generation coming from rooftop solar doubled to 14% (Clean Energy Council 2026). New South Wales (NSW) currently holds the largest installed rooftop solar capacity in the country. In February 2025, rooftop solar supplied 60% of the state's demand for the first time, which pushed grid demand down to a record low of 2,532 megawatts (MW), 484 MW below the previous record (Parkinson 2025). This event really shows the massive impact rooftop solar has on grid demand. Simshauser (2022) estimated that about 4,400 MW of installed rooftop solar could displace roughly 1,000 MW of conventional generation. This substantially changes the demand profile and puts more pressure on inflexible baseload generation as daytime demand falls.

These changes affect different people in diverse ways. In the short term, grid operators must anticipate sudden shifts in demand to keep everything running reliably as solar output fluctuates. Looking further ahead, understanding these patterns is crucial for long-term planning, such as deciding whether to build new baseload plants or fast-response peaking generators. The Federal Government also recently launched the Solar Sharer Offer (SSO), which gives NSW households three hours of free electricity during the middle of the day when there is often an oversupply. Because so much generation and consumption happens behind the meter, predicting grid demand has become much more complex. Wholesale market participants also have a financial reason to get these forecasts right, since any gap between predicted and actual demand can lead to price volatility and new trading opportunities.

Based on the available data and modelling design, this study focuses on a predictive question rather than a causal one. It examines how much recent rooftop solar data can improve half-hourly NSW demand forecasts compared to matched models that don't use PV information, and it identifies when those improvements are most significant. The main evidence comes from out-of-sample comparisons of matched models. Regression coefficients and correlations help with the interpretation, while the Australian Energy Market Operator's (AEMO) 50% probability of exceedance forecast (POE50) serves as an external benchmark.

Research questions and success criteria

Objective	Analysis	Evaluation Method
Primary Outcome	Determine whether recent rooftop PV information improves next-half-hour forecasts of NSW operational demand change compared with otherwise matched non-PV models.	Compare out-of-sample MAE and RMSE for matched PV and non-PV models evaluated on identical chronological validation windows.
Secondary Analysis 1	Determine whether the forecasting value of rooftop PV information varies across various times of day and seasons.	Compare PV-related forecast gains across time blocks and seasons using subgroup MAE and RMSE analysis.
Secondary Analysis 2	Explain why rooftop PV information contributes to forecasting performance.	Use correlation analysis, regression coefficients, cross-correlation analysis, and non-linear exploratory analysis to identify associations between rooftop PV, operational demand, and weather variables, while avoiding causal interpretation.

Since the data is observational, we reframed the original research question to focus on prediction instead of causality. Because of this, our main conclusions will depend on direct, like-for-like out-of-sample forecast comparisons.

For the project, we used half-hourly NSW demand data, rooftop PV generation, and daily weather info. We compared simple linear regression, generalised additive models, and gradient-boosted tree models using chronological validation. Throughout this process, we focused heavily on seeing how much extra predictive value the PV features add.

3 Literature Review

3.1 The Connection Between Rooftop PV and Grid Demand

Several studies have pointed out a clear connection between rooftop solar generation and a decrease in grid demand. For example, Yan and Han (2023) applied state-level regression models to half-hourly data from Australia's NEM. They found that rooftop PV output generally lowers demand during the day, which contributes to the famous duck curve effect.

Simshauser (2022) also observed that rooftop solar has significantly changed the shape of system demand, primarily by reducing minimum daytime requirements. By using a partial-equilibrium model, the study estimated that around 4,400 MW of rooftop PV in Queensland displaces about 1,000 MW of traditional generation. This dip in daytime demand puts steady pressure on inflexible baseload power. This research is particularly helpful for seeing just how large and structurally significant the link between PV and demand is on a system-wide scale.

Parkinson (2025) offers a practical example from NSW. In February 2025, rooftop solar supplied roughly 60% of the total demand for the first time, occurring alongside a record-low operational demand of 2,532 MW. This clearly shows that rooftop PV is already having a measurable impact on the NSW grid.

When you look at the regression evidence from Yan and Han (2023), the modelling by Simshauser (2022), and the NSW event noted by Parkinson (2025), a consistent pattern emerges: higher rooftop PV output goes hand in hand with lower grid demand. While it makes sense that PV physically displaces grid demand, most of the current evidence is based on observations or models rather than controlled experiments. Because of this, this report treats the relationship as a major factor but doesn't claim that the regression coefficient proves direct causality.

3.2 Weather as a Driver of PV Variability

To get a proper model of rooftop PV impact, we must account for weather, which ties directly into our second research question. For instance, Houchati, Beitelmal, and Khraisheh (2022) used support vector regression to forecast solar output by combining PV generation and consumption with weather data. They hit an accuracy rate of over 90% on sunny days, which really shows how much model performance depends on the weather. Wellby and Engerer's (2016) study is also quite relevant here since they looked at how the relationship between PV and demand shifts under different weather conditions. Their research in Australia found that large, sudden PV ramps aren't just random; they are linked to specific weather patterns. This suggests that we should include cloud cover and other weather variables when analysing PV output, and that extreme ramps might need to be handled differently than typical conditions.

Shaker et al. (2020) also point out how important weather data is when modelling rooftop PV on a regional scale. Even though individual rooftops can be unpredictable, grouping them together creates a smoother regional signal. This signal can still be forecasted using weather data and representative PV measurements. For this NSW project, which means we can combine AEMO's aggregate PV estimates with variables like cloud cover, solar exposure, and temperature. Doing this also helps explain why the connection between PV and demand might look different during clear weather compared to more variable conditions.

3.3 PV-Enhanced Modelling Relative to Non-PV Baselines

When looking at the literature relevant to the third research question, there is a consistent trend of comparing PV-enhanced models against a non-PV baseline using identical data and conditions.

For instance, Bae et al. (2022) estimated unmetered behind-the-meter PV and added it back to observed demand to reconstruct the actual load. They then used an XGBoost model to forecast that series before subtracting the estimated PV to find the expected grid demand. This method performed better than both an LSTM benchmark and an XGBoost model trained only on observed load, with the biggest gains seen during daylight hours. Even though their study focused on South Korean data, a similar strategy could likely improve demand modelling in NSW. Stratman et al. (2023) found comparable results by splitting net load into consumption and PV generation and forecasting each part separately. When evaluated in areas with high PV penetration, their method produced lower errors than models trying to forecast net load directly.

Yildiz et al. (2018) offer some helpful context regarding methodology, specifically by pointing out how much temporal resolution and volatility matter when modelling short-term electricity behaviour. They compared neural-network and regression models across various time intervals and forecast horizons. Their findings show that forecasting gets much harder when household demand is highly variable or when using exceptionally fine time increments. While they didn't model rooftop PV directly or look at how weather affects rapid PV changes, their work still supports the research questions being explored here.

Finally, even though Sinclair et al. (2026) focused on wholesale prices instead of demand, their analysis showed that forecast accuracy is heavily influenced by access to AEMO's forward-looking data. This suggests that if you compare a new model to a market operator's forecast, you might accidentally hide the actual improvements made by a new architecture or extra data. It really reinforces why it is so important to use a controlled, like-for-like comparison against a non-PV baseline trained on the exact same information.

3.4 Why PV Information is Most Valuable During Transition Periods

The fact that rooftop PV offers the biggest boost to forecast accuracy between 06:00 and 12:00 suggests we should really focus on transition periods. PV output matters just as much when it is changing rapidly and reshaping net demand as it does during peak hours. Research into renewable ramps shows that these events are different from standard forecasting scenarios because sudden weather shifts can cause abrupt changes that traditional models often miss (Zhang et al. 2025). Wellby and Engerer (2016) also connected large PV ramps in Australia to specific weather patterns. Taken together, these studies imply that the morning results are reflecting ramp information rather than just peak generation.

Behind-the-meter PV also changes how we need to look at observed demand. Bae et al. (2022) argue that unmetered PV distorts load data and that being explicit about it helps improve accuracy. This is directly relevant to the matched-model design. Even when recent demand history is a great predictor, having PV data helps us tell the difference between actual changes in consumption and changes caused by solar generation.

Flexible models are also a better fit for a PV-demand relationship that shifts throughout the day. Electricity demand isn't linear; it is influenced by a complex mix of time and weather effects. Jain and Gupta (2024) point out that these nonlinear traits can drive up forecast errors, which supports using machine-learning approaches. This justifies the use of gradient-boosted tree models and explains why a single linear coefficient just can't capture everything from morning ramps to midday stability and evening recovery.

Finally, the gap between the project model and the AEMO POE50 benchmark tells us something important. It shows that while PV provides a useful signal, an operational forecast needs a much wider range of information. PV isn't a magic fix for forecasting; instead, it is just one part of a larger system that also must account for demand momentum, weather, and overall system conditions.

3.5 Synthesis and Identified Gap

Looking at the existing literature, two main patterns stand out. First, rooftop PV is regularly linked to lower grid demand, though the coefficient in this specific project isn't enough to prove a direct cause-and-effect relationship. Second, studies that show improvements in forecasting usually compare a PV-enhanced model against a matched baseline that uses the same data but lacks PV input. This second trend gives us a solid and reproducible foundation for our revised research question.

The gap we've identified is quite specific: using recent half-hourly data from NSW, does adding explicit rooftop PV information improve forecasts for next-half-hour changes in operational demand compared to models that only use demand history and calendar info? And if it does, are those improvements mostly happening at certain times of the day or during specific seasons?

3.6 Implications for Methodology

We used half-hourly AEMO data for rooftop PV and operational demand, following the methods used by Yan and Han (2023) and Shaker et al. (2020). Through regression and correlation analyses, we looked at how PV and demand relate to one another, though we aren't claiming any direct cause and effect here.

To see how much forecasting improves, we looked at various times of day and seasons, using daily weather data to keep things in context. We tackled our main research question by comparing matched PV and non-PV models on the same sets of observations, using MAE and RMSE to measure performance. We also included a gradient-boosted tree benchmark alongside our regression models to make sure our analysis stays in line with standard forecasting practices.

4 Materials and Methods

4.1 Software

You can find everything you need to recreate this report in the Group D GitHub repository. This includes the raw and cleaned data, transformation scripts, model settings, notebooks, figures, and result tables. If you want to reproduce the specific numerical tables and figures used here, just run **scripts/run_all.py**. We have also included a lecturer-facing notebook, weekly evidence, a requirements file, and a SHA-256 manifest in the repository.

We used Python for the data preparation, exploratory analysis, and modeling. Specifically, we relied on pandas and NumPy for data management, Matplotlib for our figures, and statsmodels for HAC-adjusted OLS inference. To implement our linear, spline-additive, and histogram gradient-boosted models, we used to scikit-learn. For the final tree model, we used 150 boosting iterations, a maximum of 15 leaves, a learning rate of 0.06, a minimum leaf size of 30, L2 regularization of 1, and a random seed of 42. You can find these exact settings saved in **outputs/models/model_settings.json**.

4.2 Data Description

Table 1. Cleaned datasets and analytical coverage.

Dataset	Clean rows	Start (YYYY-MM-DD)	End (YYYY-MM-DD)	Consolidated source rows	Missing cells
Rooftop PV actual	169,438	2017-01-01	2026-09-01	182,729	48
Operational demand actual	36,528	2024-08-01	2026-09-01	0	0
Operational demand forecast	36,528	2024-08-01	2026-09-01	0	0
Weather daily	3,563	2017-01-01	2026-10-03	0	58
Common modelling dataset	36,526	2024-08-01	2026-09-01	0	1,208

This updated study looks at two connected timeframes. We use actual rooftop PV data from 2017 to 2026 to analyse long-term solar growth. When it comes to demand-related analysis and forecasting, we focus on the overlapping period between August 2024 and September 2026. This is the specific window where actual operational demand, forecasted demand, and real rooftop PV data all exist at the same time. We made this distinction on purpose so we wouldn't accidentally try to reconstruct or imply historical demand values that simply aren't available.

Our modelling data brings together AEMO NSW1 rooftop PV actuals, operational demand (both actual and forecasted), and daily Bureau of Meteorology data for things like max and min temperatures, rainfall, and solar exposure. The AEMO forecast fields include the target interval, the forecast production timestamp, and the POE10, POE50, and POE90 values. We use POE50 as our central benchmark. Since the weather data is joined by date, it offers a broad daily context rather than specific details about cloud cover throughout the day.

The modelling window runs from 1 August 2024 to 1 September 2026 and includes 36,526 aligned half-hourly rows. Because daily weather observations are applied to every half-hour within a given date, they serve as broad contextual controls instead of providing intraday cloud data. We discuss this limitation further in Section 7.2.

4.3 Data Quality

Table 2. Extended-data quality summary.

Metric	Value
Common modelling rows	36,526
Operational-demand actual rows	36,528
Operational-demand forecast rows	36,528
Common-period start	1 August 2024
Common-period end	1 September 2026
Duplicate handling	Median consolidation: source count retained
Missing weather handling	Training-set median within each model

Before we dove into the modelling, we made sure to audit the extended-data pipeline. Both the actual and forecast files for operational demand contain 36,528 clean NSW1 half-hour records, covering the period from August 2024 to September 2026. Once we aligned everything, the final modelling table ended up with 36,526 rows. While pulling the historical rooftop PV data, we noticed some duplicate region-timestamp records. We fixed this by using the median value and kept the `SOURCE_RECORD_COUNT` so we can still audit the process if needed. Most of the missing data is in the daily weather fields, but instead of just deleting those rows, we handle them during model training by using the medians from the training set.

4.4 Processing and Feature Engineering

Before we started joining everything together, we standardized all the timestamps and region codes. We also double-checked daylight-saving transitions to make sure we didn't end up with any duplicate or missing intervals. Once that was sorted, we merged the PV, demand, weather, and calendar data into a single dataset using a common half-hourly index, making sure to keep a log of any join failures or quality flags along the way.

To make our analysis and forecasting more effective, we built several lag and change features. For instance, `PV_LAG_1/2/4` and `DEMAND_LAG_1/2/4` represent values from 30, 60, and 120 minutes ago, while `DELTA_PV_MW` and `DELTA_DEMAND_MW` show the changes over the last half hour. We also included a categorical field called `PV_RAMP_DIRECTION` to track whether PV is rising, falling, or staying steady. When it came to the forecast models, we were careful to only use predictors that would be available before the target interval begins.

We defined the next-half-hour target by taking the operational demand in the following half-hour and subtracting the demand at the prediction origin. We also shifted the lagged predictors before doing the chronological split. This ensured that every row only contained information that was observable at or before its specific prediction time. This sequencing was vital because if we had calculated lags after splitting the data, or used centred transformations, we might have accidentally leaked information from the target interval into our predictors. To round things out, calendar fields captured the hour, weekday, month, and season, while the continuous lag variables helped us track the magnitude and direction of recent trends.

We chose the 30, 60, and 120-minute lags specifically to capture immediate momentum, persistence over two settlement intervals, and a broader short-term trajectory. We didn't just add longer lags for the sake of it, since adjacent half-hour observations are usually highly correlated and would just add redundancy without providing much new information. Finally, to keep the comparison fair, both the standard and PV-enhanced models used the exact same demand, calendar, and weather predictors. The only difference was that the enhanced version included PV levels and recent PV movement. This approach makes it much easier to see exactly how much value the PV data adds to the forecast.

4.5 Model Selection

We chose three different model types to find a good balance between being able to explain the results and having enough flexibility to make accurate predictions.

First, we used Multiple Linear Regression (OLS). This is great for interpretability because it gives us a direct coefficient in MW to show the relationship between PV and demand, all while we control for things like time, calendar, and weather. Next, we looked at Generalized Additive Models (GAM). These allow for smooth, non-linear relationships without us needing to define their exact shape upfront, even if they are a bit harder to interpret. Finally, we used Gradient-boosted trees. These are excellent at catching non-linear thresholds and complex interactions between demand history, PV, time, and weather, though we do lose that direct coefficient-level clarity.

We kept OLS as an interpretable benchmark, but it isn't really our main forecasting baseline. Our exploratory data analysis showed structural zeros, curved intraday profiles, and associations that change over time. Because of this, our main internal baseline is a matched gradient-boosted tree without PV, while our enhanced version adds PV levels and PV changes. We also included a spline-additive ridge model as an intermediate non-linear benchmark and used AEMO POE50 as an external benchmark.

These three model families represent a deliberate progression. OLS tests if a stable, additive relationship is enough and provides coefficients we can inspect directly. The spline-additive model relaxes that linearity requirement while keeping an additive structure, which helps with smooth intraday and weather responses. The gradient-boosted tree can handle thresholds and interactions, like how the response to a PV change might differ between the morning and the afternoon. By comparing PV and non-PV versions within each family, we can separate the question of model flexibility from the question of whether PV adds useful information.

Our primary explanatory model was:

$$\Delta Demand_t = \beta_0 + \beta_1 \Delta PV_t + \beta_2 X_t + \varepsilon_t$$

In this equation, the variables represent half-hourly changes in operational demand and rooftop PV output at time t . The control vector includes the hour, weekday, season, maximum temperature, and solar exposure, while the final term represents the model error.

4.6 Data Splits and Validation Design

The modelling table contains 36,526 common half-hour records spanning from 1 August 2024 to 1 September 2026. We sorted these by date, using the first 80% for training and setting aside the final 20% as a holdout period, which runs from 1 April 2026 at 20:00 until 1 September 2026. We applied these same holdout rows and non-PV predictors to every matched pair of with-PV and without-PV data. Just so you know, the AEMO change benchmark is only evaluated when we have a valid pre-interval POE50 forecast available.

The whole point of this matched comparison is to see exactly what PV adds to a forecast that is otherwise identical. By using the same observations and non-PV predictors, we make sure that any performance differences aren't just because of a different evaluation window or extra demand history. We also kept preprocessing and model settings consistent for every pair. You can find the tree settings in Section 4.1, and the spline transformations and ridge penalty are also important if you want to reproduce the intermediate nonlinear benchmark.

Splitting the data chronologically reflects how new observations show up in real life, but the effectiveness of this method depends on how the models were picked. If we changed features or settings after looking at

the holdout performance, then that period has already helped shape the final model and isn't truly independent anymore. That is why the extra post-selection test in Section 6.4 is so important. It helps us see if the PV gain holds up when evaluated against data that played no part in selecting the model.

We didn't use random cross-validation because it would mix up early and late observations, which could let the model accidentally learn seasonal or operational patterns from the future. Instead, the 80/20 split acts as a single forward-looking test. One downside is that the holdout period is concentrated between April and September, so it isn't seasonally balanced. We have reported the subgroup results transparently, but using a rolling-origin design across different seasonal windows would be a much stronger way to evaluate if the PV benefit stays consistent throughout the entire year.

4.7 Reproducibility and Leakage Controls

Reproducibility is important for this project since we are working with several datasets that have different coverage, time resolutions, and revision structures. To handle this, the workflow separates everything into distinct stages: raw acquisition, cleaning, feature generation, model fitting, and report outputs. Instead of relying on manual spreadsheet edits, each stage produces an auditable output. We also use fixed model settings, a random seed, common modelling rows, and exported result tables to make sure we don't accidentally mix different analytical versions when pulling figures or statistics.

Three controls were applied to prevent data leakage. First, the target is a future half-hour change, while predictor lags refer only to the prediction origin or earlier intervals. Second, missing weather values are replaced using medians from the training period rather than the full series. Third, the chronological split is completed before model fitting so that holdout data do not influence parameter estimates. These controls do not create a fully verified live forecast - publication timing and subsequent revisions are not observable - but they ensure that the offline comparison is internally consistent.

The core strategy for our predictive analysis is a matched-pair design. Within each model family, we use the exact same rows, targets, validation periods, and non-PV predictors. The only difference in the PV model is the addition of PV levels and recent PV movement. Because of this, any difference in holdout error can be directly attributed to those specific predictors rather than a different sample or a better evaluation window. It is important to note that this is a predictive attribution; it isn't a causal estimate of how demand would change if rooftop PV were physically altered.

Finally, the external AEMO comparison serves a different role. It acts as an operational reference, but we exclude any forecasts produced after the target interval. This isn't a matched ablation because AEMO uses different inputs, systems, and revision processes. Because of that, we give more weight to our internal PV versus non-PV pairs when answering the main research question, using AEMO primarily to show the remaining accuracy gap. By keeping these roles separate, we ensure that a strong operational benchmark doesn't distract from our main goal: seeing if explicit PV information adds value to our models.

4.8 Assumptions and Residual Diagnostics

We didn't expect the standard OLS assumptions of independent and homoscedastic residuals to hold up for half-hourly time-series data, so we decided to test them directly. It turns out the initial residual autocorrelation was quite high at short lags, sitting at 0.683 for 30 minutes, and it stayed significant over longer periods too, with 0.172 at two hours and 0.620 at one day. This suggests that our model wasn't quite capturing the daily patterns. Because of this, we estimated the coefficient uncertainty using HAC standard errors:

$$\hat{V}_{HAC} = (X'X)^{-1}X'\hat{\Omega}X(X'X)^{-1}$$

While using HAC standard errors helps account for uncertainty in the coefficient estimates, it doesn't fix the model's underlying structural limitations, which we discuss in Section 5.5.

4.9 Forecast Evaluation Framework

The goal of this forecasting task is to predict how demand changes. Our non-PV feature set uses things like previous demand, previous demand changes, the hour, weekday, month, and daily weather. To make it a PV-enhanced set, we also add the PV value at the time of prediction and the change in PV over the previous half hour. To keep things consistent, we calculate the AEMO forecast change by taking the POE50 demand at the target interval and subtracting the actual operational demand seen at the prediction origin. This approach is important because it prevents us from comparing demand-level errors against demand-change errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

In these equations, y_i represents the observed value, $\hat{y}_i$ is the model's prediction, and n is the total number of observations. While MAE gives a general sense of error, RMSE puts more weight on larger errors. We report both metrics in Section 6.

It is worth noting that these are rolling one-step pseudo-forecasts. This means every prediction assumes that the demand and PV values from the previous half hour were already known. Since the data doesn't show exactly when forecasts were published or updated, you shouldn't view these results as a verified measure of real-time forecasting performance.

5 Exploratory Data Analysis

5.1 Data Quality Summary

You can find a complete summary of our data quality checks in Section 4.3. Once we aligned the timestamps, the final modelling dataset consisted of 36,526 half-hourly NSW1 observations, spanning from 1 August 2024 to 1 September 2026. We only found one missing observation for Rooftop PV, and while there are a few small gaps in the daily weather variables due to station reporting issues, they are quite minor. We decided to keep all valid demand and PV intervals. For any missing weather predictors, we simply used the medians from the training set to fill in the blanks.

To break that down a bit:

- Rooftop PV only has one missing observation, which is just 0.01%.
- The daily weather fields have different amounts of missing data. For example, maximum temperature is missing only 0.25%, while sunshine hours and 3 pm cloud cover have gaps of 7.07% and 3.54%, respectively.
- These gaps are caused by occasional incomplete reporting from the stations rather than any kind of systematic error in the data.

5.2 Univariate Analysis

Figure 1. Distributions for log-transformed rooftop PV, operational demand, and next-half-hour demand change.

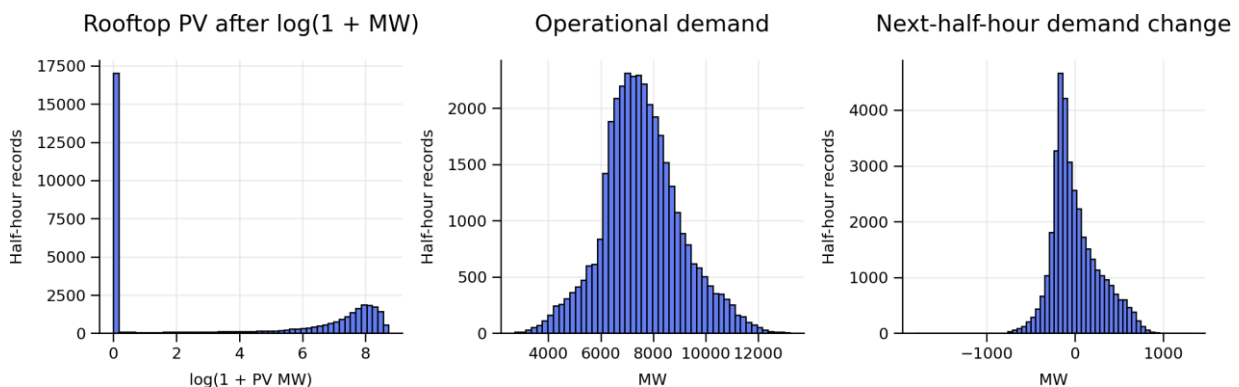


Table 3. Descriptive statistics for modelling data

Variable	N	Mean	Median	SD	Min	Max	Skew
Rooftop PV	36,526	1126.48	27.70	1558.92	0.00	6201.87	1.20
Operational demand	36,526	7568.28	7471.00	1491.38	2718.00	13204.00	0.25
Demand change	36,525	-0.03	-65.00	269.51	-1803.00	1317.00	0.69
Maximum temperature	36,430	23.80	23.40	5.20	13.60	42.70	0.54
Solar exposure	36,478	16.42	15.30	7.03	2.60	32.10	0.51

Table 3 gives a summary of the variables in their original scales, and Figure 1 displays how the main variables are distributed. You can see that rooftop PV is heavily right skewed with a skewness of 1.20. It naturally clusters at zero during the night and then stretches out into a long tail during the day. On the other hand, operational demand is more continuous and only slightly right skewed at 0.25. The half-hourly demand change stays centred around zero, but it has long tails on both the positive and negative sides, which makes sense given the typical morning and evening ramps.

Following some feedback from the lecturer, Figure 1 uses $\log(1 + PV)$ to look at the rooftop PV distribution. We added one so we could keep those zero-output periods in the data while making the positive values much easier to visualize. Because of these specific distributions, it makes sense to use both MAE and RMSE together. It also justifies why we are testing non-linear models instead of just assuming everything follows a linear-normal pattern.

5.3 Temporal Patterns

Figure 2. NSW1 daytime rooftop PV growth from 2017 to 2026; 2026 is a partial year.

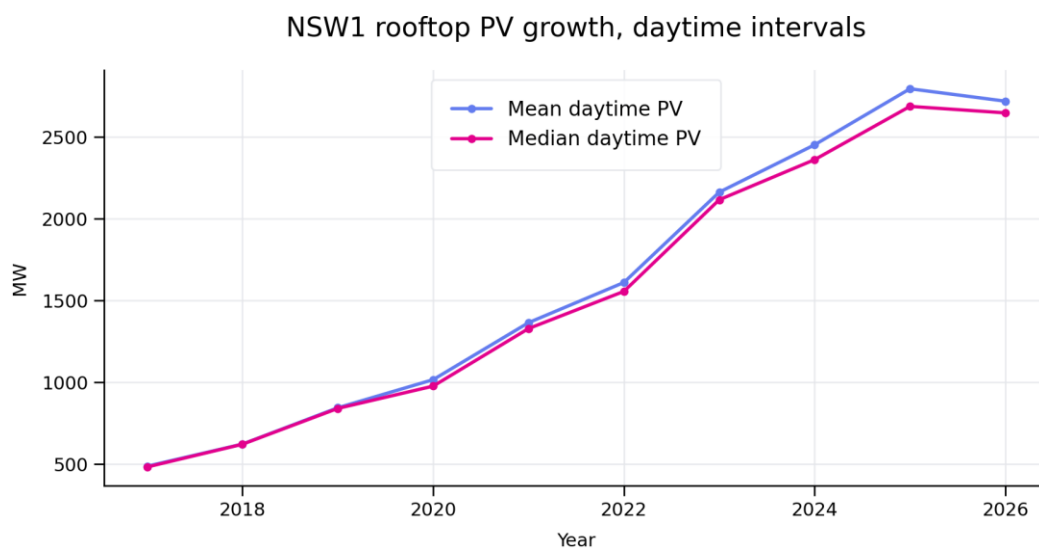
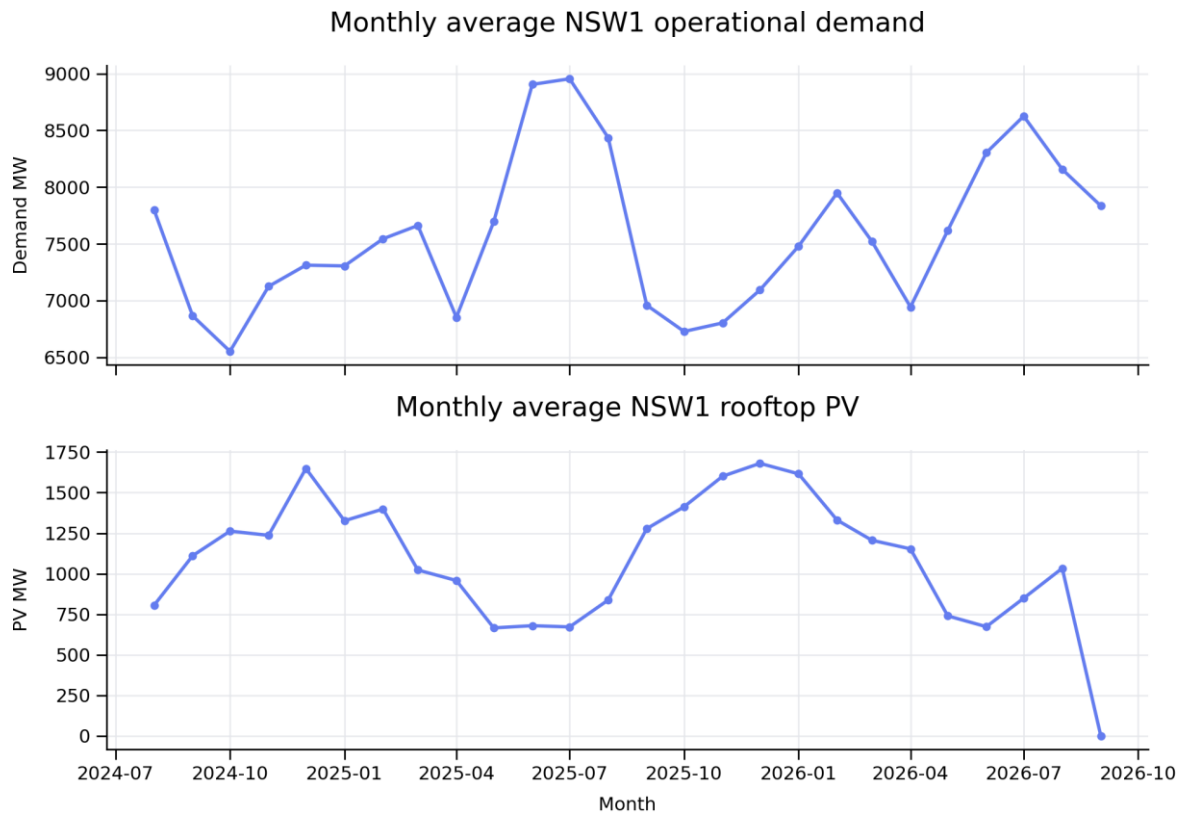


Figure 2 demonstrates rooftop PV history from 2017 to 2026, showing an almost six-fold increase in PV generation. Yan and Han (2023) attribute it to a combination of emission and climate change concerns, and consumers beginning to manage their own energy.

This long-term growth has modelling consequences. A relationship estimated when rooftop PV capacity was relatively low may not remain stable as daytime net demand is increasingly shaped by behind-the-meter generation. The level of PV in 2026 is therefore not simply a larger version of the 2017 signal: its contribution to midday demand, ramp steepness and forecast error can also change. For that reason, the extended series is used to establish structural growth, while the matched forecasting exercise is restricted to the recent common period in which demand, PV and benchmark data are aligned.

Figure 3. Monthly average NSW operational demand and rooftop PV.



If you look at Figure 3, you can see the annual cycle over the two years we modelled. Operational demand hits its peak during the winter and dips in the spring, which makes sense given that NSW's winter load is mostly driven by heating. On the other hand, rooftop PV follows a nearly opposite pattern. It peaks in late spring and summer, then drops off in the winter when the sun isn't as high in the sky.

Figure 4. Seasonal half-hour profiles for rooftop PV and operational demand.

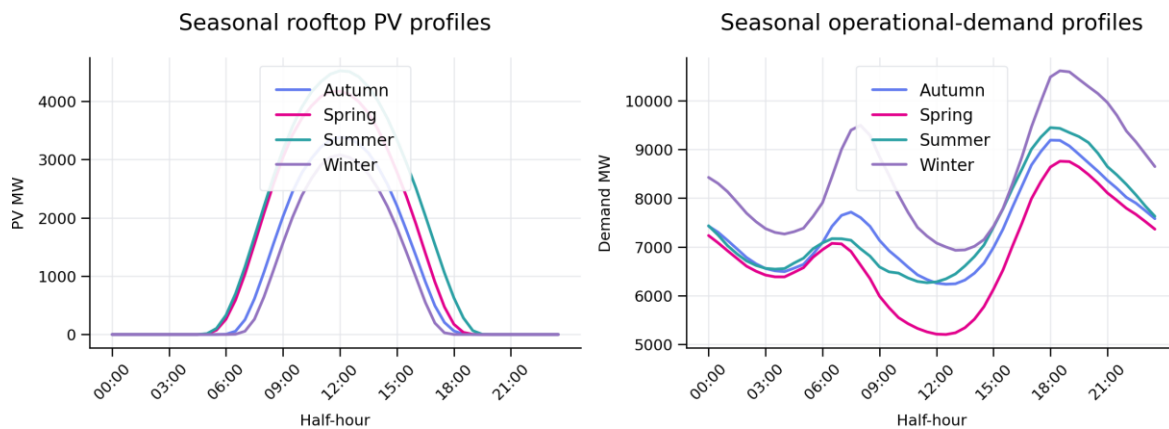


Figure 4 illustrates how both series look throughout the day depending on the season. PV follows a bell-shaped curve that centres around midday, hitting peaks of roughly 4,500 MW in the summer and about 3,000 MW in the winter. Demand, on the other hand, shows that classic duck-curve pattern. You can see a morning peak, a midday dip that gets deeper as PV output climbs, and then a sharp evening ramp starting around 15:00. For instance, in the spring, demand drops to about 5,200 MW at midday before climbing to 8,700 MW by 18:30. That is a massive ramp of 3,500 MW in just six hours. These transition periods, when PV is either ramping up in the morning or fading out in the evening, are especially hard to predict. This is exactly why we want to see if including PV data adds value during these specific times.

The seasonal profiles also explain why PV and demand are roughly inverse but not perfect mirror images. Operational demand is influenced by a lot of different things, like heating, cooling, business hours, and daily household routines, alongside rooftop generation. In the winter, shorter days mean less PV, while heating needs drive up the load in the morning and evening. In the summer, stronger and longer solar production pushes grid demand down around midday, but cooling needs can keep demand high. Because these varied factors overlap, the impact of any specific PV level changes depending on the hour and the season, rather than just acting as a constant subtraction from total demand.

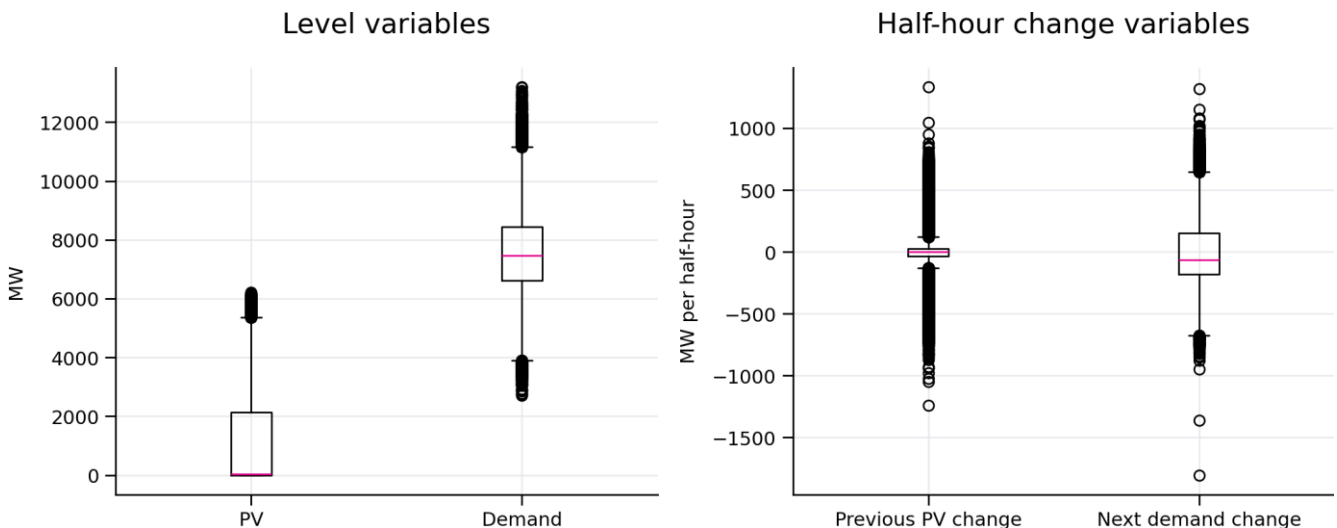
When you look at Figures 2 through 4 together, the relationship between PV and demand shifts based on the year, the season, and the time of day. Because of this, a single stationary linear model just wouldn't be enough to create a complete forecasting model.

5.4 Outlier Analysis

Table 4. Tukey 1.5 × IQR screening.

Variable	Lower fence	Upper fence	Flagged N	Flagged %
ROOFTOP_PV_MW	-3212.17	5353.61	417	1.1%
OPERATIONAL_DEMAND_MW	3901.00	11157.00	784	2.1%
TARGET_DEMAND_CHANGE_MW	-674.00	646.00	864	2.4%

Figure 5. Original-scale boxplots for level and half-hour change variables.

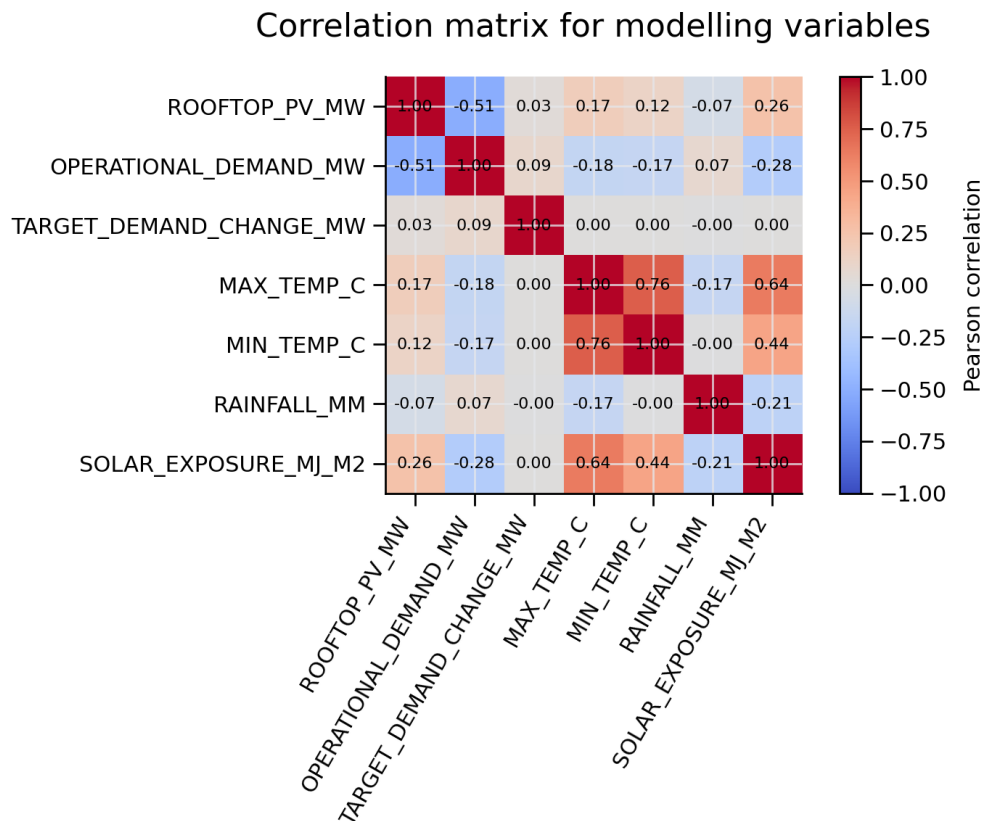


We used Tukey's 1.5 x IQR rule to spot potential outliers, then double-checked them against their timestamps and nearby intervals. Looking at Table 4 and Figure 5, it's clear that most of the flagged data points matched asymmetric distributions or real half-hour ramps instead of being errors. It is important to remember that a statistical outlier isn't always an invalid observation. In this case, those quick changes in the morning and evening are vital events that a forecasting model needs to learn. If we just removed them, we'd narrow the target distribution, which might make our validation accuracy look better on paper but would hurt performance during the times that matter most.

The Tukey rule isn't perfect for rooftop PV either, mostly because the data has a huge mass of zeros overnight and a completely different range during the day. The fences it creates describe the whole combined distribution rather than what we'd expect for a specific hour or season. Because of this, high daytime PV readings can get flagged even when they're perfectly normal. By keeping these observations, we preserve the real operating range. Plus, by reporting both MAE and RMSE, we can show both the typical absolute error and how much the larger misses impact the results.

5.5 Bivariate and Multivariate Analysis

Figure 6. Correlation matrix for common-period modelling variables.



As shown in Figure 6, there is a moderate negative correlation between rooftop PV and operational demand ($r = -0.51$), which lines up with the typical duck-curve mechanism. We also see a strong positive correlation between maximum and minimum temperatures ($r = 0.76$). Both temperature metrics are positively linked to solar exposure ($r = 0.64$ and 0.44 , respectively), which makes sense given seasonal and daylight patterns. Interestingly, each weather variable has only a weak linear relationship with either PV output (r from -0.07 to 0.26) or operational demand (r from -0.28 to 0.07), though solar exposure does show the strongest connection in both instances. Finally, the forecast target, which is the change in demand for the next half hour, is basically uncorrelated with the other variables, including PV.

Figure 7. Daily mean rooftop PV against solar exposure, and daily mean operational demand against maximum temperature.

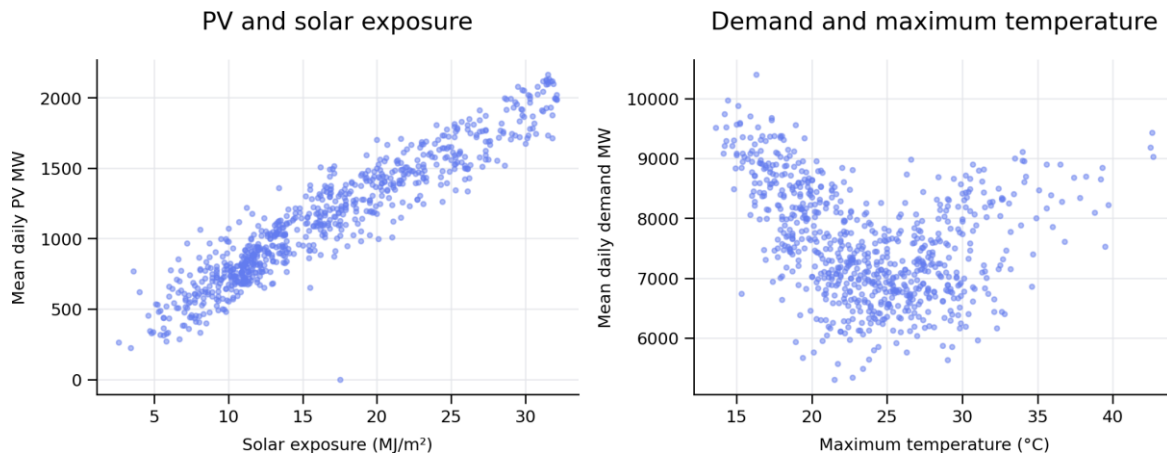
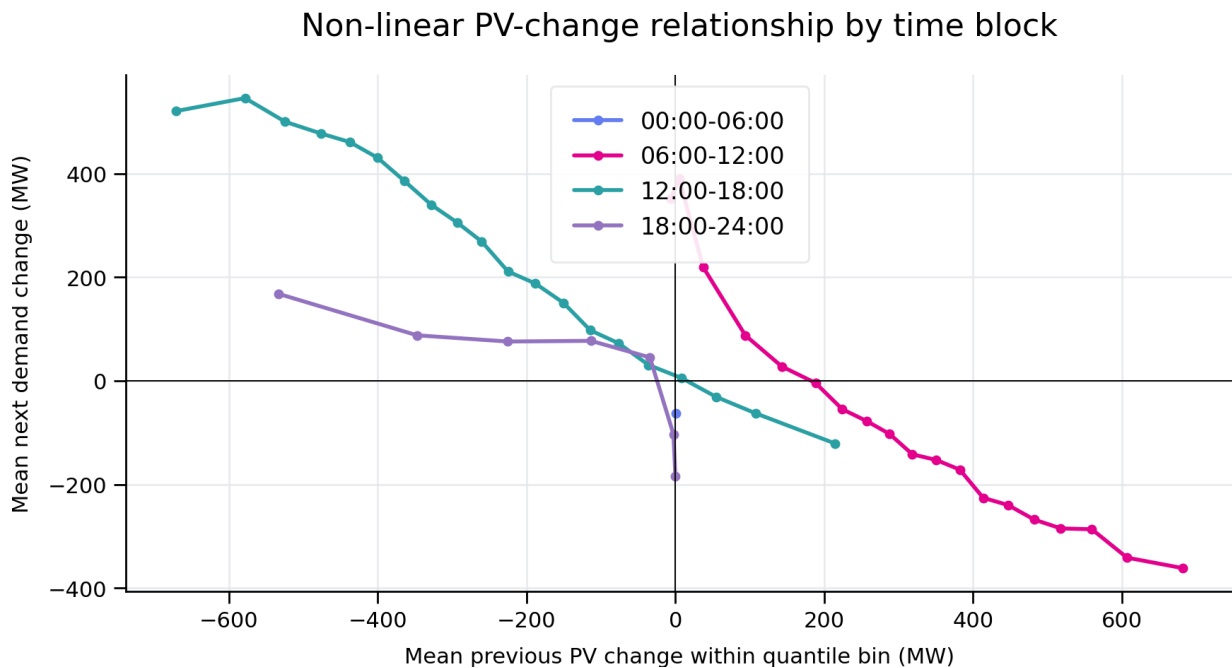


Figure 7, Demand and maximum temperature, helps explain why the weather correlations are weak. Daily mean PV rises almost linearly with solar exposure, but daily mean demand has a U-shaped relationship with maximum temperature. Demand is highest on the coldest days (below 17°C) and rises again above 30°C, with a minimum between 22°C and 27°C, because both heating and cooling add load.

Figure 7, PV and solar exposure, shows that daily PV increases with solar exposure. These patterns support the use of flexible models and highlight the limitation of applying one daily station value to every half-hour.

Figure 8. Binned relationship between previous PV change and next-half-hour demand change by time block.



While the Pearson correlation matrix shows a linear relationship, it doesn't really capture the full nonlinear pattern. To get a better look, Figure 8 groups previous PV changes into bins and plots the average demand change for the next half hour by time block. Since the shape and slope change depending on the PV range and the time of day, we found that splines and tree models work much better than OLS for our primary forecasting.

- **Morning (06:00 to 12:00):** The average demand change drops from roughly +400 MW when the previous PV change is near zero to about -360 MW once PV rises by around 680 MW.
- **Afternoon (12:00 to 18:00):** Demand can jump by as much as 500 MW following large PV drops, but it only declines slightly when PV rises.
- **Evening (18:00 to 24:00):** The reaction to a PV drop is a bit smaller here, sitting at around +170 MW.
- **Overnight:** PV change is basically zero, so the data stays concentrated at a single point.

Figure 9. Cross-correlation between previous PV change and future operational-demand change.

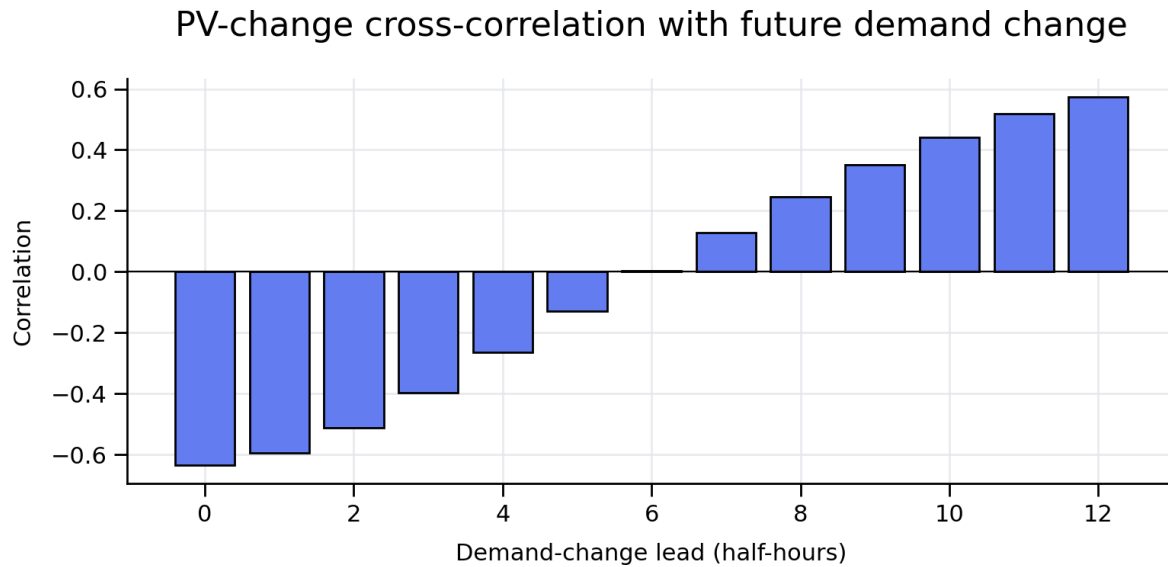


Figure 9 illustrates how changes in previous PV levels correlate with future demand changes. At lag zero, there is a correlation of -0.63, but this weakens over time until it hits zero at the six-period mark, which represents three hours. After that, the correlation actually turns positive, peaking at around +0.57 by the twelve-half-hour mark. This shift is likely just a reflection of the natural daily cycle rather than a delayed effect from PV, so it shouldn't be mistaken for a causal relationship. Because the association is so strong in the short term and drops off so quickly, it makes sense to focus primarily on the very next half-hour.

5.6 Implications of EDA for Model Specification

Based on the EDA, there are three main reasons why we expect PV to offer incremental value that changes over time. To start, PV levels are heavily influenced by daylight and the seasons, which means a lot of its data overlaps with standard calendar variables. Second, recent changes in PV show a much stronger link to future demand shifts during the morning and afternoon transitions compared to what happens overnight. Finally, the way this relationship behaves changes depending on the scale and direction of the PV movement, so it is not just a simple, constant linear effect.

These insights shaped how we designed our matched models. We kept the demand history to capture recent net-load momentum, while using the hour, weekday, month, and weather to provide the necessary recurring context. The PV-enhanced models then add just the recent PV levels and their movement. We used OLS to test for an average additive association, a spline model to capture smooth curvature, and a boosted tree to let PV interact with both time and the current state of demand. Also, the decision to evaluate subgroups by time block and season was based directly on the patterns seen in Figures 2 through 9, rather than being something we added after seeing the forecasting results.

6 Analysis of Results and Discussion

6.1 Explanatory Association and Baseline Design

We kept OLS as an explanatory model and a linear forecasting benchmark, even though it wasn't the primary baseline for measuring the specific value added by PV. For the internal forecasting baseline, we used a gradient-boosted tree without PV, meaning the version with PV only adds information specific to solar power to that same structure. We also used AEMO POE50 as an external operational benchmark. Finally, we tested a spline-additive model to see if smooth nonlinearity could perform well on its own, without the complex interactions that tree-based models typically capture.

Table 5. Selected PV coefficients from the explanatory OLS model with HAC standard errors.

Explanatory term	Coefficient	HAC SE	95% lower	95% upper	p-value
PV level at origin	0.012	0.001	0.011	0.014	1.77e-71
Previous PV change	-0.211	0.005	-0.221	-0.202	0

When we expanded our explanatory models by adding more predictors, it became obvious that including a demand lag term mattered much more than adding a PV lag. In fact, adding a PV lag across various seasonal periods didn't really change the validation MAE in any meaningful way. However, adding a demand change-lag dropped the error significantly, bringing the mean validation MAE down from 93.59 MW to 73.65 MW. This suggests that recent demand momentum is a far stronger predictor of current movements than the history of rooftop PV.

The distinction between association and forecasting value helps explain why this happens. While there is a negative correlation between PV and operational demand, that just describes how the two series move together over time. It doesn't measure how much PV improves a forecast once you already know the recent demand and calendar patterns. Similarly, while short-lead cross-correlation shows a link between recent PV changes and subsequent demand, some of that information might already be captured by other predictors.

Recent demand is especially useful because it already reflects the combined impact of electricity use, rooftop solar, and other factors affecting grid demand. Using the previous demand change also gives the model a baseline to estimate whether demand is trending up or down. Because of this, PV needs to offer something extra beyond what the recent demand pattern already tells us to improve the forecast. This lines up with what we saw in the explanatory model comparison, where adding demand momentum provided a huge boost, while PV only provided a smaller incremental gain in the final matched trees.

We should keep this context in mind when looking at the 1.5% reduction in tree MAE. The real question isn't whether rooftop PV has a meaningful relationship with grid demand, but whether measuring it separately helps predict the next half-hour after all other useful data is already in the model. The holdout results show there is a modest extra benefit. This also explains why the exploratory data analysis can show clear PV-related patterns even if those patterns don't lead to a massive drop in forecast error.

6.2 When Is PV Information Most Useful?

Comparing the matched trees confirms that the value of PV isn't constant throughout the day. The biggest improvement happens between 06:00 and 12:00, when the MAE drops by 2.01 MW. This doesn't contradict the fact that PV levels peak around midday. It's important to remember that the forecast value measures the extra information PV provides beyond just demand history and calendar patterns, not the raw amount of solar generation itself. In the morning, both PV and demand are changing quickly, so recent PV movement can help tell the difference between periods where rising solar is offsetting load growth. We saw gains of 0.05 MW overnight, 0.37 MW between 12:00 and 18:00, and 1.33 MW between 18:00 and 24:00. While direct error comparisons, binned nonlinear analysis, and short-lead cross-correlation support this, the relationship remains predictive rather than causal.

Formal statistical testing for these subgroup differences was a bit more limited. An interaction-term OLS specification used in previous studies showed that the afternoon-period interaction coefficient (+0.0526) had a confidence interval that crossed zero (-0.0006 to +0.1058). Even though the solar-exposure interaction was statistically significant (+0.0213 per 10 MJ/m², with an interval of +0.0010 to +0.0415), the effect was quite small. The evidence for the second research objective is much clearer at the descriptive and forecasting levels, where correlations and PV gains show a consistent time-of-day pattern, rather than through formal interaction significance in a single linear model. I've noted this distinction explicitly to avoid overstating the result as a confirmed statistical interaction.

The difference between morning and afternoon performance suggests that high PV generation isn't the only thing that makes the information useful. In the morning, solar output is climbing just as electricity use starts to shift with the start of the day. Recent PV movement might help the model distinguish between a rise in actual consumption and a drop in the demand supplied by the grid. Around midday, solar output can be extremely high without offering much new information beyond what's already in the demand history and regular daily patterns. This is a plausible explanation for the smaller afternoon gains, even if it isn't a mechanism directly established by the model comparison.

The improvement seen in the evening offers another perspective. As solar production drops, a larger portion of electricity use must come from the grid, alongside the usual evening shifts in consumption. Recent PV observations might help the model recognize this transition. The near-zero gain overnight also fits this idea: when PV output is basically sitting at zero, its recent levels and movement don't really offer much extra info about what the next demand change will be.

6.2.1 Additional Insights from the Final Validation Analysis

Table 6. Summary of additional findings from the final analysis.

Finding	Evidence from final analysis	Interpretation
Morning transition	PV gain = 2.01 MW MAE	Largest value occurs before the solar peak.
Evening handover	PV gain = 1.33 MW MAE	Recent PV helps identify the transition as solar fades.
Demand momentum	93.59 to 73.65 MW after adding demand-change lag	Demand history dominates, while PV remains incremental.
Stability	Bootstrap mean gain = 0.88 MW; 95% interval = 0.19-1.57 MW	Small but consistent average benefit.
Benchmark gap	AEMO 58.21 MW vs PV tree 62.13 MW	PV is useful but not sufficient for operational parity.

We can answer the research question through five key findings. First, PV information is most helpful during transition periods. Second, demand momentum remains the primary predictor for short-term horizons. Third, while the PV gain is modest, it stays stable day to day. Finally, the AEMO benchmark suggests an operational information gap still exists. You can find a summary of this evidence in Table 6.

6.3 Primary Research Question: Does PV Improve a Matched Non-linear Forecast?

Table 7. Authoritative chronological-holdout model comparison.

Model	MAE (MW)	RMSE (MW)	R ²	Test N
AEMO POE50 change benchmark	58.21	80.51	0.915	7,305
Tree with PV	62.13	83.45	0.909	7,305
Tree without PV	63.07	84.94	0.905	7,305
Spline with PV	80.52	105.32	0.855	7,305
Spline without PV	86.20	114.15	0.829	7,305
OLS with PV	89.24	117.30	0.820	7,305
OLS without PV	95.89	126.57	0.790	7,305
Zero-change persistence	219.80	276.18	-0.000	7,305

We reran the comparison to see how persistence, OLS, spline-additive, and gradient-boosted models performed on the same chronological holdout. The tree model that included PV reached an MAE of 62.13 MW and an RMSE of 83.45 MW. In comparison, the matched tree without PV had an MAE of 63.07 MW and an RMSE of 84.94 MW. This means adding PV led to an incremental improvement of 0.94 MW in MAE, which is about 1.5%, along with a 1.49 MW drop in RMSE. Even so, the AEMO POE50 was still the strongest benchmark with an MAE of 58.21 MW. These results suggest that PV does improve an otherwise matched nonlinear model, but they don't prove that PV will improve every single forecast or that this specific association is causal.

6.3.1 Magnitude of PV benefit across model families

As you can see from the results in Table 7, the impact of adding PV varies depending on which model you use. For the OLS model, adding PV cut the MAE by 6.65 MW, while the spline-additive model saw a reduction of 5.68 MW, and the tree model improved by 0.94 MW. When you compare these to their respective versions without PV, it works out to improvements of roughly 6.9%, 6.6%, and 1.5%.

Table 8. Comparison of PV improvement between model types.

Model	Without PV MAE	With PV MAE	MAE gain
OLS	95.89 MW	89.24 MW	6.65 MW
Spline additive	86.20 MW	80.52 MW	5.68 MW
Gradient boosted tree	63.07 MW	62.13 MW	0.94 MW

Even though the simpler models saw a bigger boost from adding PV, the tree model still gave the most accurate project forecasts. In fact, even without PV, its MAE was lower than what the OLS or spline-additive models achieved with PV included. One reason for this might be that the tree already picks up on so much of the daily demand pattern through demand history and calendar variables that there isn't much left for PV to add. The comparison seems to back this up, though it doesn't tell us exactly which relationships the tree has learned.

The tree model with PV still sat 3.92 MW above AEMO's aligned MAE. This helps put things in perspective: while recent PV data did improve the project model, it wasn't enough to close the gap with the external benchmark. AEMO works well as a reference point for forecasting, but the matched project models give us much more direct evidence of the actual value PV adds.

Looking at the success criteria, the main goal was met in a narrow but consistent way. Both MAE and RMSE improved when PV was added to the matched tree using the same holdout rows. The secondary analysis regarding the time of day also holds up, since most of the gains happened in the morning and, to a slightly lesser degree, the evening. However, the seasonal conclusions aren't as strong because the holdout data is imbalanced and only includes one observation at the spring boundary. Finally, while the binned and cross-correlation patterns support the explanatory objective, those analyses show predictive associations rather than proving a direct causal link.

6.3.2 Integrated Interpretation of the Evidence

The results are most convincing when you look at all the separate analyses together. First, the exploratory data analysis shows that PV follows specific hourly and seasonal patterns, and that recent PV shifts have a nonlinear relationship with upcoming demand changes. We then used a matched forecasting comparison to see if this data adds value once you've already accounted for recent demand, the calendar, and the weather. Finally, the time-block and bootstrap analyses help pinpoint exactly where these gains happen and if they are spread out across the days we studied. While no single piece of evidence tells the whole story, all three points toward the same conclusion.

The comparison between different model families adds some important context. PV reduces error by 6.65 MW in the OLS model and 5.68 MW in the spline-additive model, but only by 0.94 MW in the tree model. This doesn't mean PV isn't important in the best-performing model. It's more than a flexible tree can already notice parts of the daily pattern using just recent demand and calendar variables, which leaves less new information for the PV predictors to provide. Even so, the tree with PV still has the lowest error of all the models tested. This shows that model flexibility and PV data work well together, but the extra benefit you get from PV depends on what your baseline model can already figure out on its own.

The subgroup results help explain why the average gain seems a bit modest. PV doesn't really add much information overnight since its levels and changes are usually zero. Its biggest contribution happens during the morning ramp, with a smaller benefit during the evening transition. When you average these different periods over an entire day, the gains seen during those rapid change intervals get diluted. For anyone using this in practice, it suggests that you shouldn't judge a predictor's value based solely on its overall average. Looking at conditional performance can show you specific times of day when that information is much more valuable for making decisions.

The positive daily bootstrap interval makes the results more dependable because it makes it less likely that the improvement was just caused by a few isolated half-hour windows. That said, this interval only describes uncertainty based on our specific day-resampling method, so we shouldn't assume it applies to every future season or system condition. Since summer wasn't included in the final holdout and we only had one spring boundary record, our evidence regarding the time of day is stronger than our evidence regarding the season. The most honest conclusion is that recent PV provides a small average benefit throughout the period we studied, with much clearer value during daily transitions, though we still need to test its stability year-round.

6.4 Residual Diagnostics and Model Limitations

Figure 13. Paired daily bootstrap distribution of the tree-model MAE gain from PV.

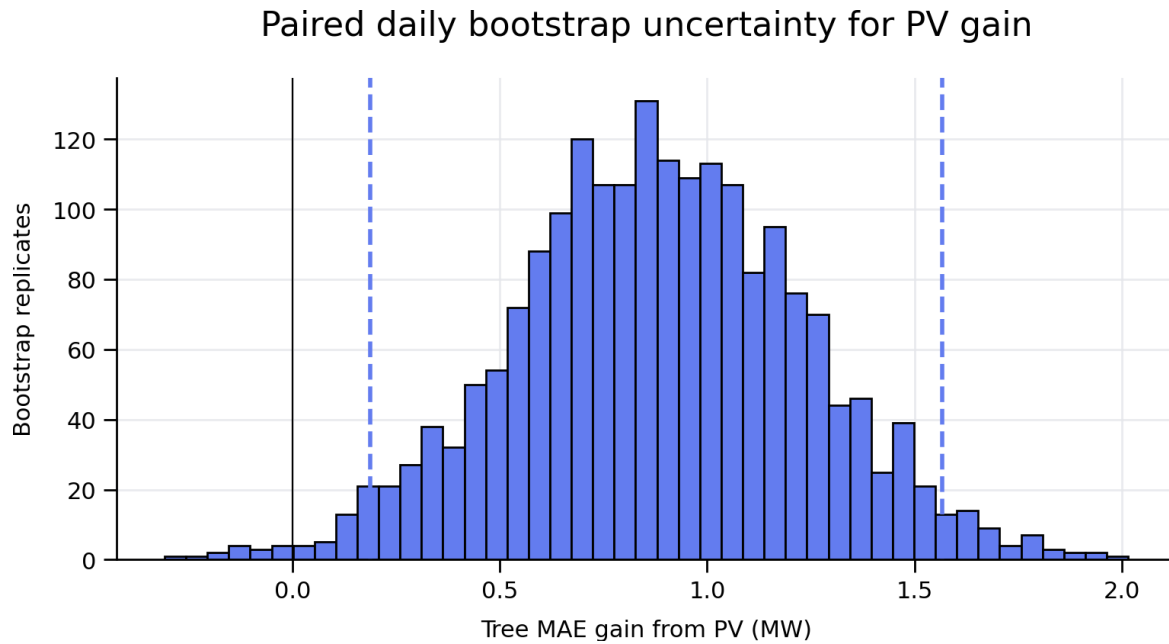


Figure 13 shows whether the improvements in the matched tree are actually stable or if they're just being driven by a few specific time periods. We ran 2,000 resamples of entire days and saw an average MAE gain of 0.88 MW, with a 95% percentile interval ranging from 0.19 to 1.57 MW. Because this interval stays above zero, it suggests the average gain is consistent, even if the actual improvement is a bit modest.

When we checked the residuals from the explanatory OLS model, we noticed significant autocorrelation. Adding demand history helped a little, but not entirely. Before we included the demand lag, the residual correlation was 0.683 at 30 minutes and 0.620 at one day. After adding the lag, the 30-minute correlation dropped to 0.193, but the one-day correlation remained high at 0.425. This means the model still isn't fully capturing the regular daily cycle, so we need to keep that in mind when looking at the coefficient uncertainty estimates, which use HAC standard errors.

The main limitations are:

- The evaluation uses rolling one-step pseudo-forecasts and assumes that demand and PV values from the previous half-hour were already available. Since we don't have exact publication and revision timestamps, these results shouldn't be taken as verified real-time performance.
- Weather observations come from Sydney Airport as a proxy for NSW, so they might not catch local variations in cloud cover or solar exposure.
- The PV lag terms showed substantial multicollinearity, with variance inflation factors around 11.8 for the middle lags. We kept a minimal lag structure to make sure the model stays easy to interpret.
- The holdout period is mostly concentrated in autumn and winter, has only one observation at the spring boundary, and leaves out summer entirely. Also, the selected model hasn't been tested on a fully independent post-selection period.

Figure 14. Holdout MAE for OLS, spline-additive and tree models with PV by season.

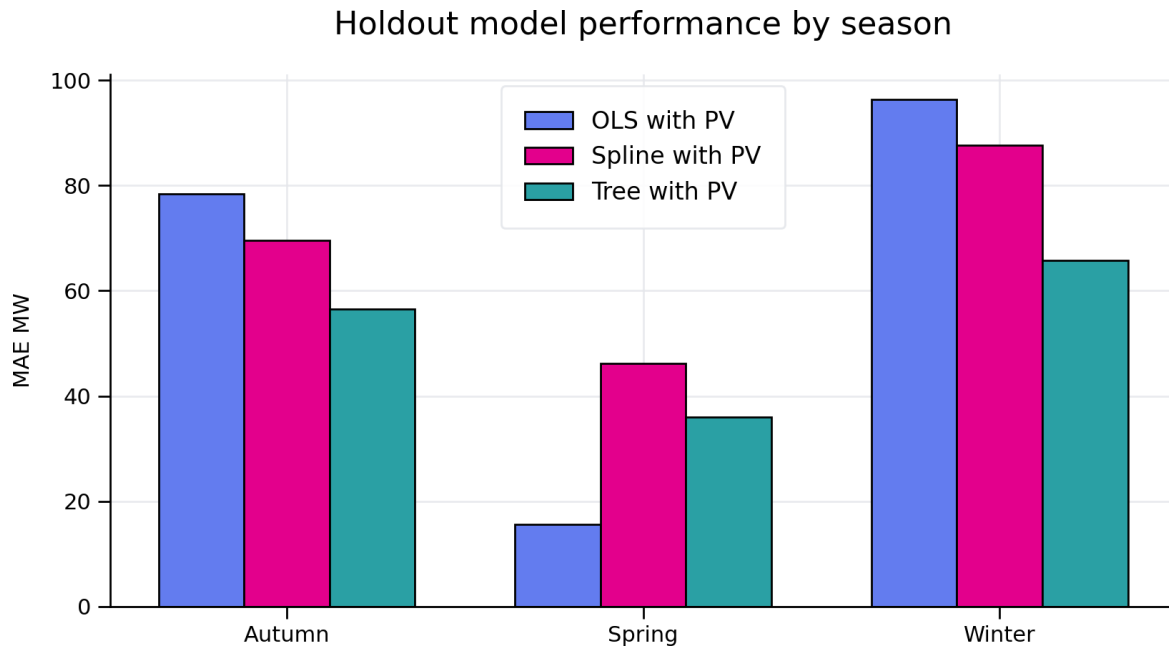


Figure 14 compares the OLS, spline-additive and tree models by season on the final holdout. The tree model performs best in the autumn and winter groups, supporting the choice of a nonlinear model. The single spring boundary record is retained for completeness but is not interpreted as seasonal evidence. PV gains were 1.67 MW in autumn and 0.46 MW in winter, showing that the magnitude varies by season.

The paired comparison is important because both models forecast the same intervals. Resampling the same days for each model allows the analysis to compare their errors under the same conditions, while retaining the dependence between half-hourly observations within a day. The positive interval therefore supports an average improvement across the evaluated days; it does not mean that PV improves every individual forecast or performs equally well in every season.

There are limits to this uncertainty estimate. Sampling individual days does not retain dependence between consecutive days, which may share weather conditions or demand patterns. Resampling blocks of consecutive days would provide a useful sensitivity check. The reported bootstrap mean of 0.88 MW is also slightly below the full-holdout gain of 0.94 MW. These estimates summarise the comparison in different ways, and their difference should be checked against the treatment and weighting of days before drawing further conclusions.

Figure 15. Residual distribution and residual sequence for the PV-enhanced tree.

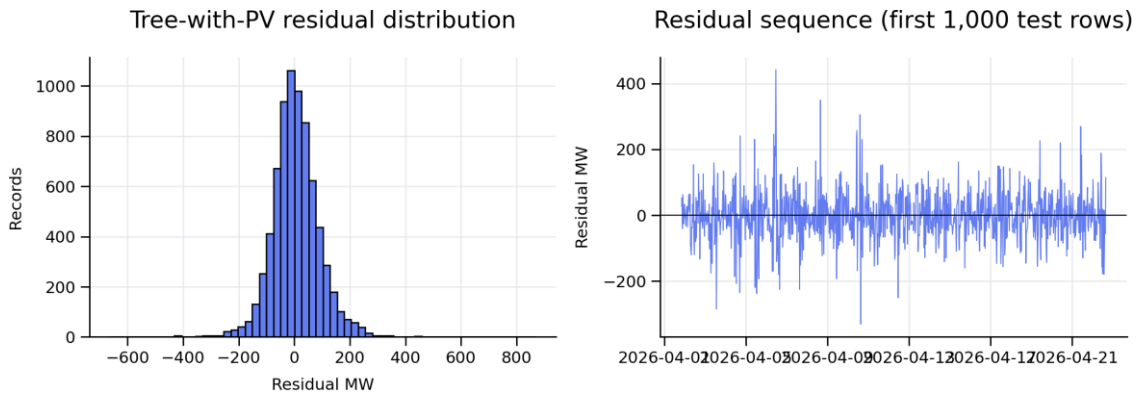
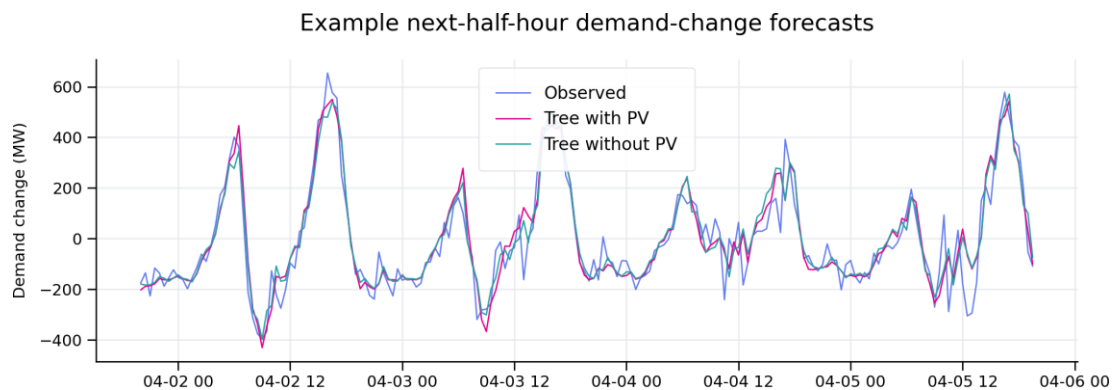


Figure 15 shows that the PV-enhanced tree residuals remain centred broadly around zero but have long tails and visible clustering over time. The tail behaviour indicates that a small number of intervals still produce errors much larger than the typical forecast miss, while the sequential pattern indicates that adjacent errors are not independent. This matters because summary metrics alone can conceal periods when the model repeatedly under- or over-predicts. It also supports the use of chronological testing and a paired day-level bootstrap rather than uncertainty calculations that assume independent and identically distributed residuals.

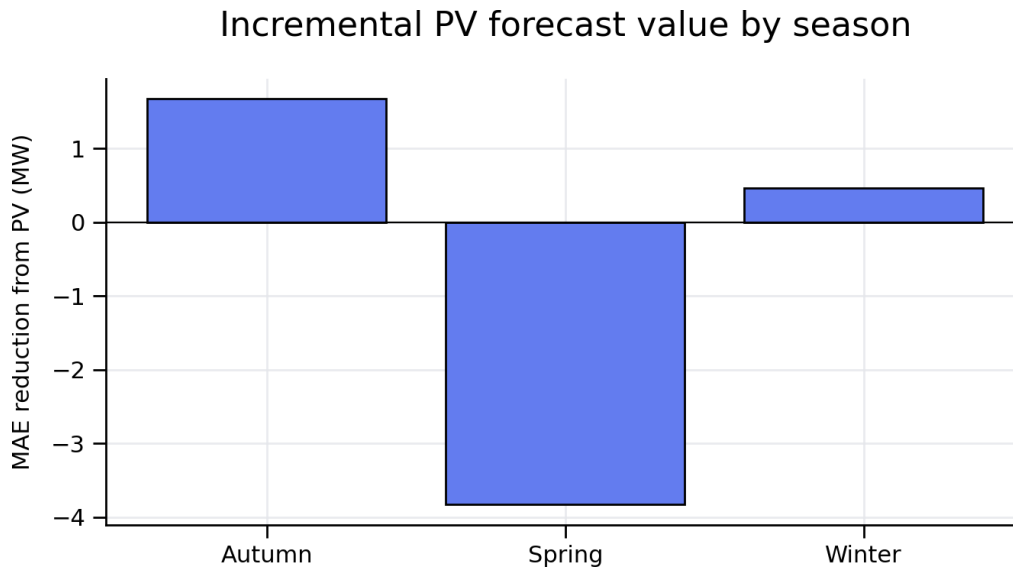
Figure 16. Observed and matched tree demand-change forecasts over four holdout days.



The forecast example is a great way to intuitively check if the model is following the main ramps and reversals. You can see the differences between the two tree versions most clearly when there are sharper transitions.

Because of this, Figure 16 should be viewed as a diagnostic tool rather than definitive proof of general performance. While both trees follow the broad direction of the observed ramps, they tend to smooth out some sharp reversals and can occasionally miss the full magnitude of extreme changes. The gap between the two predictions is most obvious around transition periods, which lines up with the time-block results. In areas where the lines overlap, the demand history and calendar structure already account for most of the movement. Where they diverge, PV is shifting the model's estimate, though that change isn't always an improvement for every single interval.

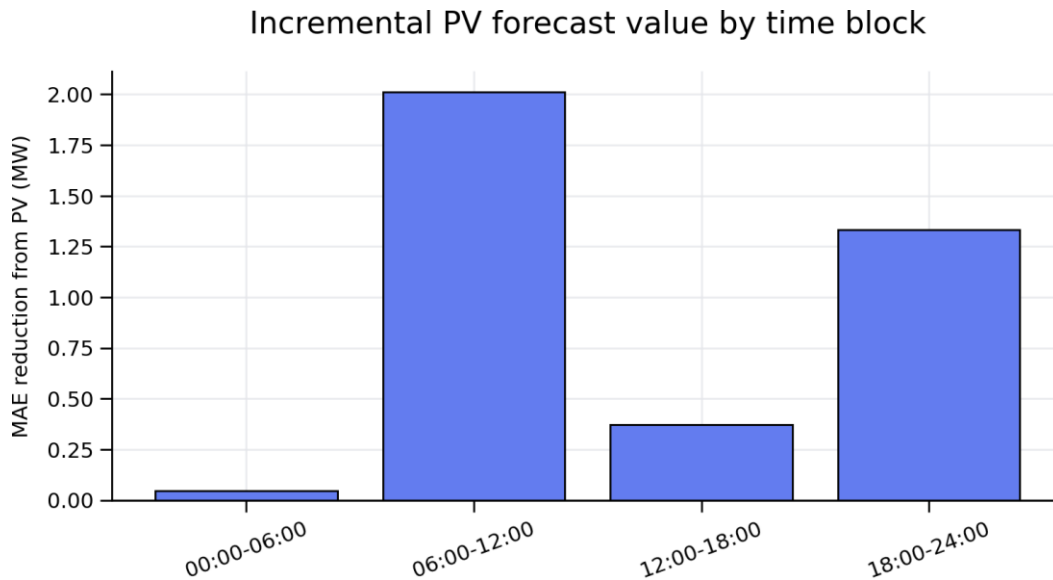
Figure 17. Incremental PV value by season on the chronological holdout.



Looking at seasonal gains helps us figure out if the overall results are just concentrated in one specific time of year. The data from autumn and winter is quite useful, but since we only have a single record for the spring boundary, we can't really draw any conclusions from it.

It is also important to look at the seasonal bars in the context of their sample sizes. We have enough holdout observations in autumn and winter to see some descriptive patterns, but we can't claim a stable annual ranking yet. This is because summer is missing from the data and spring is barely represented. The bigger gains seen in autumn might just be due to more intense changes in weather and daylight as we head into winter, but the current setup doesn't let us distinguish that from the way the holdout data is composed. We really need to use rolling seasonal validation before we can say for sure that PV is more valuable in one season over another.

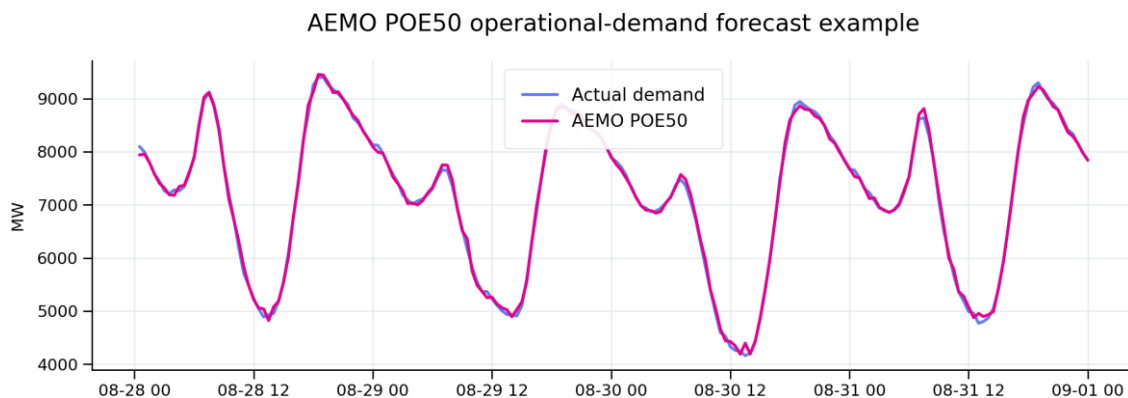
Figure 18. Reduction in matched tree MAE from adding PV by time block.



The direct comparison between the with-PV and without-PV models answers the main question by time of day. The biggest improvement happens during the morning solar ramp, not at the peak PV level.

Figure 18 also clarifies the practical meaning of the aggregate 0.94 MW gain. The overall average combines intervals in which PV is highly informative with intervals in which it is almost constant. Overnight, the 0.05 MW change is negligible because PV is normally zero. The 2.01 MW morning improvement is the clearest evidence that recent solar movement adds information during a rapid transition, while the 1.33 MW evening improvement suggests value as PV output declines. The smaller afternoon gain indicates that high solar output by itself is less useful once regular intraday patterns and recent demand are already known.

Figure 19. Representative actual operational demand and AEMO POE50 level forecast.



The level-forecast example confirms that AEMO follows the general demand profile and provides context for the aligned change benchmark. We report level errors separately from demand-change errors.

Figure 19 is not a second evaluation target. It shows the demand level so that the benchmark can be inspected in its original form, whereas Table 7 evaluates the change from the observed prediction origin to the target interval. Keeping these views separate avoids treating a level forecast error as directly comparable with a demand-change error. The level plot is useful for identifying broad tracking and timing, but the aligned change calculation is the relevant basis for the reported benchmark MAE and RMSE.

Figure 20. AEMO forecast horizon after excluding forecasts produced after the target interval.
AEMO operational-demand forecast horizon

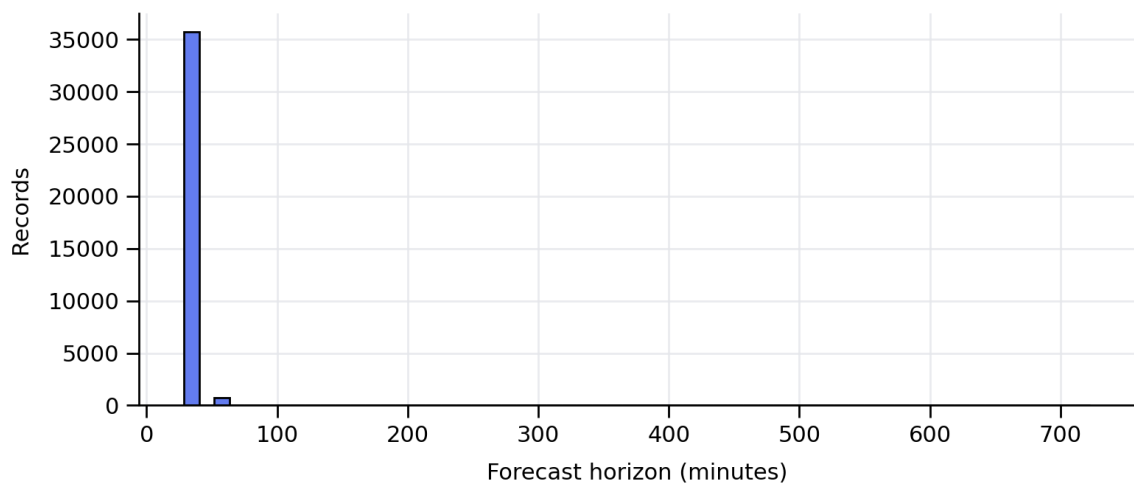
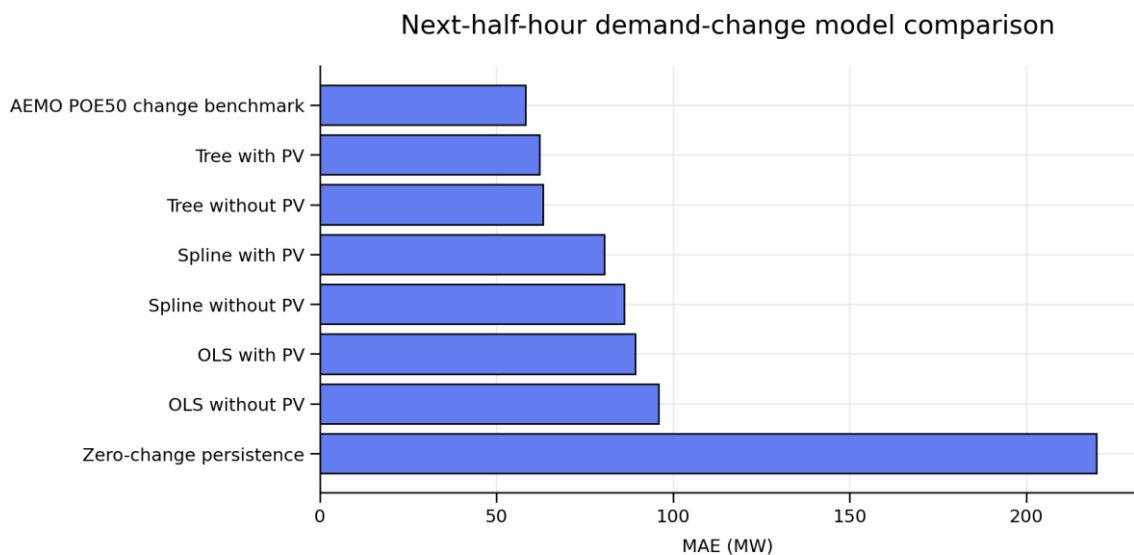


Figure 20 validates the timing condition used for the external comparison. Forecasts produced after their target interval were excluded, and the retained positive horizons indicate that the forecast was available before the interval being evaluated. Even so, the distribution shows that the AEMO forecasts were not necessarily produced at a single uniform lead time. The benchmark should therefore be interpreted as an operational reference assembled from valid pre-interval forecasts, not as a perfectly matched model trained and issued under the same information conditions as the project models.

Figure 21. Chronological holdout MAE comparison across project models, persistence and aligned AEMO POE50.



The main benchmark chart compares all models against the same target for demand changes over the next half hour. It shows that our flexible project models have an edge over linear and persistence baselines, though AEMO POE50 remains the strongest external comparison.

Looking at Figure 21, we can draw two distinct conclusions. First, nonlinear structure is important, as the tree models perform better than the spline, OLS, and zero-change persistence options. Second, adding PV provides incremental value within every model family, even if that gain shrinks as the baseline model gets more flexible. AEMO's top result doesn't change the validity of our matched comparison because the two are answering different questions. The matched pairs show exactly what PV adds to the project pipeline, while the AEMO result shows the remaining gap between our approach and a broader operational forecasting process.

7 Conclusion and Further Issues

To ensure our final comparison was as accurate as possible, we used a regenerated extended-data pipeline and set aside a chronological holdout period from 1 April 2026 at 20:00 until 1 September 2026. Out of everything we tested, AEMO POE50 performed the best, hitting the lowest aligned demand-change MAE at 58.21 MW. When we looked at the specific project models, the tree model that included PV showed an MAE of 62.13 MW and an RMSE of 83.45 MW. By comparison, the matched tree without PV had an MAE of 63.07 MW and an RMSE of 84.94 MW. Basically, adding PV reduced the tree MAE by 0.94 MW, which is a 1.5% improvement. We also ran a paired daily bootstrap, which estimated a mean gain of 0.88 MW, with a 95% percentile interval between 0.19 and 1.57 MW. Just so you know, these updated results replace all the figures found in previous drafts and reports.

7.1 Practical Implications

The modeling suggests that PV data is most effective during periods of rapid solar changes rather than being helpful throughout the entire day. Because of this, short-term demand forecasting should use recent PV measurements to supplement historical demand and calendar predictors, particularly in models capable of handling non-linear interactions. Since the improvement is relatively small, anyone considering implementation should weigh the data and maintenance requirements against the actual value of a slight but steady reduction in error.

The scale of this improvement matters when looking at its practical use. Reducing the MAE from 63.07 MW to 62.13 MW means the average absolute error was 0.94 MW lower across the intervals we evaluated. This doesn't mean every single prediction improved by that much. Some forecasts might have improved significantly, while others may have changed very little or even become less accurate. The time-of-day results explain why it is important to look at the overall average alongside the specific conditions where PV was most useful.

RMSE also dropped from 84.94 MW to 83.45 MW. Since RMSE puts more weight on large errors, it offers a different perspective on performance rather than just relying on MAE. The fact that both measures improved supports the finding that PV helped the matched tree, though these aggregate results don't prove it reduced the most extreme errors or improved every major ramp event. The positive daily bootstrap interval provides more evidence of an average benefit under the chosen resampling method, but it doesn't change the fact that the overall gain is small.

AEMO's lower MAE serves as a useful benchmark for the limits of this project's model. It shows that adding PV data didn't bring the model up to the level of the external benchmark, even though it did improve the internal comparison. Therefore, recent PV should be seen as just one useful input within a much broader forecasting process. The results don't pinpoint exactly which additional information or modeling choices give AEMO the advantage, so we can't attribute that gap to one specific missing feature.

Whether this improvement is worth implementing depends on the specific setting. For an organization that already collects timely PV data, getting that extra forecasting benefit might be fairly easy. However, if that information requires new collection processes, quality checks, or ongoing maintenance, the small average gain needs to justify those extra steps. This project hasn't estimated financial savings, scheduling benefits, or implementation costs. Its main contribution is showing that recent PV information can improve short-horizon forecasts, especially during daily transitions, while leaving the decision of whether to deploy it to a separate operational assessment.

7.2 Limitations and Future Work

There are five main limitations to these findings:

- First, using daily weather data from just one station means we can't really capture how clouds move throughout the day across NSW.
- Second, while we have rooftop PV history going back quite a way, the consistent demand data doesn't go as far back, so we can't estimate long-run PV-demand relationships before August 2024.
- Third, because the source workbooks didn't clearly show when revisions were made, we had to use the median to consolidate repeated timestamps.
- Fourth, the chronological holdout is a bit heavy on autumn and winter, which creates a seasonal imbalance.
- Finally, the model we chose hasn't been tested on a completely independent post-selection period yet.

Moving forward, it would be best to prioritize sub-daily weather data from multiple spots, get better metadata on AEMO revisions, and use rolling seasonal validation windows.

Beyond these technical points, it is worth noting that this analysis looks strictly at rooftop PV, even though other trends are changing grid demand. For example, home battery adoption is really picking up speed. Under the Federal Government's Cheaper Home Batteries Program, 500,000 batteries were installed in only one year (Albanese 2026). Since people can use stored energy during morning and evening peaks, this could help flatten demand. On the other side, AEMO expects data centre electricity use to jump from 3% to 13% over the next ten years, which is about the same as the total household demand in NSW and Victoria combined (Evans 2026). The Federal Government has proposed laws that would require data centres to fund offsetting renewable energy, but it is still hard to say how much this will impact net-demand growth. In the end, forecasting methods will just have to keep evolving as technology, habits, and regulations change.

7.3 Prioritised Validation Roadmap

First, we really need a genuine post-selection evaluation period. This means all feature choices, transformations, and model settings must be frozen before we process any new observations. Running the matched comparison on this untouched data would be the best way to see if that 0.94 MW MAE gain holds up outside the data used to build the workflow. This is much more useful than just tuning the model against the current holdout since constant adjustments would eventually ruin that holdout's independence.

Second, we should implement rolling-origin validation across full seasonal windows. Instead of relying on a single April-to-September split, using a sequence of training and test periods would let us compare PV gains across summer, autumn, winter, and spring with enough data to be meaningful. We should report results both overall and by time blocks within each season. This will help us figure out if the morning advantage is a stable trend or just a byproduct of the specific weather and daylight patterns in our current holdout.

Third, we need better weather and forecast-timing data. Using sub-daily observations from several locations across NSW would represent cloud movement much better than just relying on one daily value from Sydney Airport. It would also be helpful to keep track of forecast issue times and revision histories. This way, we can evaluate every predictor and benchmark using only the information that was available at a specific point in time. These steps would shift the analysis from a consistent pseudo-forecast to something much closer to a real-world operational simulation.

Finally, the focus should be on targeted robustness analysis rather than just adding more complexity to the model. Useful checks would include resampling blocks of consecutive days, comparing performance during high-ramp versus low-ramp intervals, and seeing if error reductions are tied to just a few specific events. We should also monitor how performance changes as installed PV capacity grows. All of these tests need to keep the matched design intact so that we are always measuring the actual value of PV, rather than just seeing differences caused by different samples or model settings.

7.4 Final Conclusion

The data shows exactly how much recent rooftop solar information helps when forecasting demand changes for the next half hour. This solar data is most effective during daily transition periods. Specifically, it offers the biggest boost during the morning solar ramp and a slightly smaller benefit as production drops off in the evening. Even though recent demand momentum is still the main predictor for short-term changes, side-by-side comparisons show that adding PV data provides a small but steady improvement. For instance, including PV data lowered the tree model's MAE from 63.07 MW to 62.13 MW, compared to the AEMO POE50 benchmark of 58.21 MW. Because of this, PV information should be viewed as just one part of a larger forecasting system rather than a standalone fix. These findings show that solar data adds incremental value depending on the time of day, but they don't prove causality or suggest that existing forecasting processes should be replaced.

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